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Time taken	1 hour 24 mins
Grade	47.00 out of 100.00

Question **1**

Correct

Mark 1.00 out of 1.00

12. Depth first search always expands the ____ node in the problem representation of a search tree.

Select one:

- ☐ a. Minimum cost
- ☐ b. Shallowest
- ☐ c. Child node
- ☒ d. Deepest ✓

The correct answer is: Deepest

Question **2**

Incorrect

Mark 0.00 out of 1.00

7. What kind of transmission medium is most appropriate to carry data in a computer network that is exposed to electrical interferences?

Select one:

- ☐ a. Microwave
- ☒ b. Optical fiber ✗
- ☐ c. Unshielded twisted pair
- ☐ d. Coaxial cable

The correct answer is: Coaxial cable

Question **3**

Incorrect

Mark 0.00 out of
1.00

91. Which of the following is a correct syntax to create an object of interface named Animal

Select one:

- ☒ a. `Animal animal = new Animal ();` ❌
- ☐ b. `new Animal ();`
- ☐ c. It is not supported to create an object of Animal
- ☐ d. `Interface Animal animal = new Animal ();`

The correct answer is: It is not supported to create an object of Animal

Question **4**

Correct

Mark 1.00 out of
1.00

86. Identify one which is preemption based algorithm when scheduling CPU:

Select one:

- ☒ a. RR ✔
- ☐ b. SJF
- ☐ c. Priority
- ☐ d. FCFS

The correct answer is: RR

Question **5**

Correct

Mark 1.00 out of
1.00

What is the output of the following java program?

```
class Flower {  
    public Flower () {  
        this ("Rose ");  
        System.out.print("Flower. ");  
    }  
    public Flower (String name) {  
        System.out.print("I am "+name);  
    }  
}  
  
public class RoseTest extends Flower {  
    public RoseTest (){  
        super ();  
        System.out.print("Hi there! ");  
    }  
    public static void main(String [] args) {  
        Flower flower = new RoseTest();  
    }  
}
```

Select one:

- ☐ a. Hi there! Rose flower I am.
- ☒ b. I am Rose Flower. Hi there! ✓
- ☐ c. I am Flower. Rose! Hi there!
- ☐ d. Hi there! I am Rose Flower.

Your answer is correct.

The correct answer is: I am Rose Flower. Hi there!

Question **6**

Correct

Mark 1.00 out of
1.00

10. Which of the following is true about MAC address?

Select one:

- ☐ a. It is also known as logical address
- ☐ b. It is used for subnetting purpose
- ☐ c. It is represented by 128 bits
- ☒ d. It is commonly assigned by the manufacturer of the NIC ✓

The correct answer is: It is commonly assigned by the manufacturer of the NIC

Question **7**

Correct

Mark 1.00 out of 1.00

1. One of the following is true about lexical and syntax analyzers

Select one:

- ☒ a. Scanner is a co-routine of parser ✓
- ☐ b. Lexical analyzer works on simple recursive constructs a language
- ☐ c. Syntax analyzer works on simple none recursive constructs a language
- ☐ d. Syntax analyzer is a co-routine of lexical analyzer

The correct answer is: Scanner is a co-routine of parser

Question **8**

Incorrect

Mark 0.00 out of 1.00

26. Which of the following is the least strong security encryption standard?

Select one:

- ☐ a. WPA
- ☐ b. WEP
- ☐ c. WPA3
- ☒ d. WPA2 ✗

The correct answer is: WEP

Question **9**

Correct

Mark 1.00 out of 1.00

79. Which component of Active Directory provides a searchable catalog of objects across multiple domains in a forest?

Select one:

- ☐ a. Domain
- ☐ b. Domain controller
- ☒ c. Global Catalog ✓
- ☐ d. Sites

The correct answer is: Global Catalog

Question **10**

Incorrect

Mark 0.00 out of
1.00

92. Consider a banking application that needs to store customer information such as name, address, and account balance. A Customer class that contains private fields for these attributes is created, as well as getter and setter methods to access and modify those fields. This allows us to keep the data private and enforce business logic within the class, while still allowing other parts of the program to interact with the object through well-defined interfaces. Which one of the following object-oriented concepts will fit with this application?

Select one:

- ☐ a. Inheritance
- ☐ b. Encapsulation
- ☐ c. Polymorphism
- ☒ d. Abstraction ✖

The correct answer is: Encapsulation

Question **11**

Correct

Mark 1.00 out of
1.00

2. Type checking is one of the tasks performed by a compiler during

Select one:

- ☒ a. Semantic analysis ✔
- ☐ b. Intermediate code generation
- ☐ c. Lexical analysis
- ☐ d. Syntax analysis

The correct answer is: Semantic analysis

Question **12**

Correct

Mark 1.00 out of
1.00

20. Precedence of regular expression in decreasing order is

Select one:

- ☐ a. + , a , *
- ☐ b. . , * , +
- ☒ c. * , . , + ✔
- ☐ d. . , + , *

The correct answer is: * , . , +

Question **13**

Correct

Mark 1.00 out of 1.00

16. The first step of machine learning is_____.

Select one:

- ☐ a. Testing
- ☐ b. Modeling
- ☒ c. Data acquisition ✓
- ☐ d. Data representation

The correct answer is: Data acquisition

Question **14**

Incorrect

Mark 0.00 out of 1.00

4. Assume that production $T \rightarrow T * F$, provided that T , $*$, and F are grammar symbols, is a final item of a given LR(0) item. How many grammar symbols will be popped from the parser stack?

Select one:

- ☐ a. 6 grammar symbols
- ☐ b. 2 grammar symbols
- ☐ c. 1 grammar symbols
- ☒ d. 3 grammar symbols ✗

The correct answer is: 6 grammar symbols

Question **15**

Incorrect

Mark 0.00 out of 1.00

51. All of the following are examples of integral data types EXCEPT _____.

Select one:

- ☐ a. int
- ☐ b. short
- ☒ c. char ✗
- ☐ d. double

The correct answer is: double

Question **16**

Incorrect

Mark 0.00 out of
1.00

89. A process stack does not contain one of the following, which one?

Select one:

- ☐ a. PID of child process
- ☐ b. Local variables
- ☒ c. Return addresses ✖
- ☐ d. Function parameters

The correct answer is: PID of child process

Question **17**

Incorrect

Mark 0.00 out of
1.00

57. A web based application model that used to add, modify and access objects hieratically.

Select one:

- ☐ a. RMI
- ☐ b. DOM
- ☒ c. MVC ✖
- ☐ d. RPC

The correct answer is: DOM

Question **18**

Incorrect

Mark 0.00 out of
1.00

85. One of the operating system securities that identify each user of the system and associate the executing programs with those users. Which one?

Select one:

- ☒ a. Separation ✖
- ☐ b. Confidentiality
- ☐ c. Authentication
- ☐ d. Integrity

The correct answer is: Authentication

Question **19**

Correct

Mark 1.00 out of 1.00

8. Which one of the following applications of network is used to assign IP address to hosts automatically?

Select one:

- ☐ a. Mailing system
- ☒ b. DHCP ✓
- ☐ c. File transfer protocol
- ☐ d. DNS

The correct answer is: DHCP

Question **20**

Correct

Mark 1.00 out of 1.00

59. A Javascript popup box used to alert dialog displaying the text "Welcome JS" is created by calling _____.

Select one:

- ☐ a. `alert.window( "Welcome JS" );`
- ☐ b. `window( "<alert> Welcome JS </alert>";`
- ☐ c. `window( "alert = \"Welcome JS\" );`
- ☒ d. `window.alert( "Welcome JS" );` ✓

The correct answer is: `window.alert( "Welcome JS" );`

Question **21**

Incorrect

Mark 0.00 out of 1.00

68. The amortized time complexity of the dynamic array problem is?

Select one:

- ☒ a. $O(\log n)$ ✗
- ☐ b. $O(n \log n)$
- ☐ c. $O(1)$
- ☐ d. $O(n)$

The correct answer is: $O(1)$

Question **22**

Correct

Mark 1.00 out of 1.00

27. Which of the following is not a cybercrime?

Select one:

- ☐ a. Malware
- ☐ b. Denial of Service
- ☒ c. AES ✓
- ☐ d. Man in the Middle

The correct answer is: AES

Question **23**

Correct

Mark 1.00 out of 1.00

6. Which of the following is not essential component of a data communications system?

Select one:

- ☒ a. Administrator ✓
- ☐ b. Transmission media
- ☐ c. Sender
- ☐ d. Protocol

The correct answer is: Administrator

Question **24**

Correct

Mark 1.00 out of 1.00

The relation book (title, price) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list?

```
SELECT title
FROM book as B
WHERE (SELECT count(*)
FROM book as T
WHERE T.price>B.price) < 5
```

Select one:

- ☒ a. Titles of the five most expensive books ✓
- ☐ b. Title of the fifth most inexpensive book
- ☐ c. Titles of the four most expensive books
- ☐ d. Title of the fifth most expensive book

Your answer is correct.

The correct answer is: Titles of the five most expensive books

Question **25**

Correct

Mark 1.00 out of 1.00

55. _____ allow you to specify the style of your page elements (spacing, margins, etc.) separately from the structure of your document.

Select one:

- ☐ a. Session
- ☐ b. Style declarations
- ☐ c. Text Editors
- ☒ d. Cascading Style Sheets ✓

The correct answer is: Cascading Style Sheets

Question **26**

Correct

Mark 1.00 out of 1.00

74. What are the advantages of arrays?

Select one:

- ☐ a. Elements in an array cannot be sorted
- ☐ b. Index of first element of an array is 1
- ☐ c. Objects of mixed data types can be stored
- ☒ d. Easier to store elements of same data type ✓

The correct answer is: Easier to store elements of same data type

Question **27**

Incorrect

Mark 0.00 out of 1.00

$(a+b)^*$ is equivalent to

Select one:

- ☐ a. b^*a^*
- ☒ b. $(a^*b^*)^*$ ✗
- ☐ c. a^*b^+
- ☐ d. a^*b^*

Your answer is incorrect.

The correct answer is: b^*a^*

Question **28**

Incorrect

Mark 0.00 out of 1.00

53. Which of the following JavaScript cannot do?

Select one:

- ☐ a. Handle to events, errors and exceptions
- ☒ b. to validate data ✖
- ☐ c. to access database data and file system
- ☐ d. to manipulate HTML elements

The correct answer is: to access database data and file system

Question **29**

Incorrect

Mark 0.00 out of 1.00

46. The expression `static_cast(6.9) + static_cast(7.9)` evaluates to ____.

Select one:

- ☐ a. 14
- ☒ b. 14.8 ✖
- ☐ c. 15
- ☐ d. 13

The correct answer is: 13

Question **30**

Incorrect

Mark 0.00 out of 1.00

How can the `index.php` script access the email form element of the following HTML form?

```
<form action="index.php" method="post">  
<input type="text" name="email"/> </form>
```

Select one:

- ☐ a. `$_POST['text']`
- ☒ b. `$_GET['email']` ✖
- ☐ c. `$_SESSION['text']`
- ☐ d. `$_POST['email']`

Your answer is incorrect.

The correct answer is: `$_POST['email']`

Question **31**

Incorrect

Mark 0.00 out of
1.00

82. Which of the following is a primary function of a domain controller in an Active Directory environment?

Select one:

- ☒ a. User management ✖
- ☐ b. File sharing
- ☐ c. Searching files
- ☐ d. Authentication

The correct answer is: Authentication

Question **32**

Correct

Mark 1.00 out of
1.00

31. One is not correct about binary number system.

Select one:

- ☐ a. The symbols used are 0 and 1
- ☐ b. It positional weighted number
- ☐ c. it also used as a machine language
- ☒ d. The base is 3 ✔

The correct answer is: The base is 3

Question **33**

Incorrect

Mark 0.00 out of
1.00

23. Which of the following act violates cyber security?

Select one:

- ☐ a. Threat
- ☐ b. Exploit
- ☐ c. Attack
- ☒ d. Vulnerability ✖

The correct answer is: Attack

Question **34**

Correct

Mark 1.00 out of 1.00

35. Amongst the ACID properties of a transaction, the 'Durability' property requires that the changes made to the database by a successful transaction persist

Select one:

- ☒ a. Always, even if there is a failure of any kind ✓
- ☐ b. Except in case of a disk crash
- ☐ c. Except in case of an operating system crash
- ☐ d. Except in case of a power failure

The correct answer is: Always, even if there is a failure of any kind

Question **35**

Correct

Mark 1.00 out of 1.00

78. Declares the host that's the most authoritative for the zone and, as such, is the best source of DNS information for the zone.

Select one:

- ☐ a. DNS Start of Authority Record
- ☒ b. DNS A Record ✓
- ☐ c. DNS NS Record
- ☐ d. DNS PTR Record

The correct answer is: DNS A Record

Question **36**

Incorrect

Mark 0.00 out of 1.00

42. What is the best data type definition for Oracle when a field is alphanumeric and has a fixed length?

Select one:

- ☐ a. LONG
- ☐ b. CHAR
- ☒ c. VARCHAR2 ✗
- ☐ d. NUMBER

The correct answer is: CHAR

Question **37**

Correct

Mark 1.00 out of 1.00

83. Which command-line tool is used to display the current TCP/IP network configuration on a Windows Server?

Select one:

- ☐ a. Ping
- ☐ b. Tracert
- ☐ c. Nslookup
- ☒ d. Ipconfig ✓

The correct answer is: Ipconfig

Question **38**

Correct

Mark 1.00 out of 1.00

60. Which of the following is a correct order of activities in system development life cycle?

Select one:

- ☐ a. Requirement Collection => Analysis => Design => Coding=> Testing=> Maintenance =>Deployment
- ☒ b. Requirement Collection => Analysis => Design => Coding=> Testing=>Deployment=> Maintenance ✓
- ☐ c. Requirement Collection => Design=> Analysis => Coding=> Testing=> Maintenance =>Deployment
- ☐ d. Requirement Collection => Analysis => Coding =>Design => Testing=>Deployment=>Maintenance

The correct answer is: Requirement Collection => Analysis => Design => Coding=> Testing=>Deployment=> Maintenance

Question **39**

Incorrect

Mark 0.00 out of 1.00

72. The postfix form of the expression $(A + B) * (C * D - E) * F / G$ is?

Select one:

- ☒ a. $AB + CDE * - * F * G /$ ✗
- ☐ b. $AB + CD * E - FG / **$
- ☐ c. $AB + CD * E - F ** G /$
- ☐ d. $AB + CD * E - * F * G /$

The correct answer is: $AB + CD * E - * F * G /$

Question **40**

Incorrect

Mark 0.00 out of 1.00

58. Which of the following is not correct about PHP?

Select one:

- ☐ a. Restrict users to access some pages of your website
- ☐ b. Encrypt and decrypt data using built-in algorithms
- ☐ c. Handle sessions of pages using setCookie() function
- ☒ d. Enable and disable cookies using setCookie() function ✖

The correct answer is: Handle sessions of pages using setCookie() function

Question **41**

Correct

Mark 1.00 out of 1.00

47. Which keyword is used for structure definition?

Select one:

- ☐ a. None of them
- ☐ b. def_struct
- ☐ c. structure_def
- ☒ d. struct ✔

The correct answer is: struct

Question **42**

Incorrect

Mark 0.00 out of 1.00

52. An/A _____ is ideal when the style is applied to many pages.

Select one:

- ☐ a. style sheet
- ☐ b. cascading style sheet
- ☒ c. internal style sheet ✖
- ☐ d. external

The correct answer is: external

Question **43**

Correct

Mark 1.00 out of 1.00

75. The post-order traversal of a binary tree is O P Q R S T. Then possible pre-order traversal will be

Select one:

- ☐ a. T Q O S P R
- ☒ b. T Q O P S R ✓
- ☐ c. T Q R S O P
- ☐ d. T O Q R P S

The correct answer is: T Q O P S R

Question **44**

Correct

Mark 1.00 out of 1.00

22. Regular expression are

Select one:

- ☐ a. Type 1 language
- ☐ b. Type 2 language
- ☒ c. Type 3 language ✓
- ☐ d. Type 0 language

The correct answer is: Type 3 language

Question **45**

Correct

Mark 1.00 out of 1.00

77. . Which of the following is an infix expression?

Select one:

- ☐ a. +ab
- ☐ b. abc+*
- ☒ c. (a+b)*(c+d) ✓
- ☐ d. ab+c*

The correct answer is: (a+b)*(c+d)

Question **46**

Incorrect

Mark 0.00 out of 1.00

73. The data structure required to check whether an expression contains a balanced parenthesis is?

Select one:

- ☐ a. Stack
- ☒ b. Array ✖
- ☐ c. Queue
- ☐ d. Tree

The correct answer is: Stack

Question **47**

Incorrect

Mark 0.00 out of 1.00

38. What is the impact of not including a JOIN command when using multiple tables in a query?

Select one:

- ☐ a. The query will not work – an error is generated
- ☐ b. All records in each table are associated with all records in the other tables
- ☐ c. The foreign keys in each table are linked to the primary keys in the other tables
- ☒ d. The primary keys in each table are joined together ✖

The correct answer is: All records in each table are associated with all records in the other tables

Question **48**

Incorrect

Mark 0.00 out of 1.00

25. Which of the following DDoS in mobile systems wait for the owner to trigger the cyber attack?

Select one:

- ☐ a. Programs
- ☐ b. Virus
- ☐ c. Botnets
- ☒ d. Worms ✖

The correct answer is: Botnets

Question **49**

Correct

Mark 1.00 out of 1.00

3. Which one of the following is not true about stack-based shift-reduce parser?

Select one:

- ☒ a. Initially the stack contains \$ and starting none terminal of the grammar ✓
- ☐ b. The parsing is successful if the stack is left with \$ and the starting symbol of the grammar and the input buffer is left with \$
- ☐ c. \$ is used to mark the bottom of the stack and the right end of the input buffer
- ☐ d. The parser repeats shifting and reducing actions until the stack is empty or an error is happened

The correct answer is: Initially the stack contains \$ and starting none terminal of the grammar

Question **50**

Incorrect

Mark 0.00 out of 1.00

44. ____ limits who gains access to the database while ____ limits what a user can access within the database.

Select one:

- ☐ a. Access authentication, view definition
- ☐ b. Data access, user monitoring
- ☒ c. Access authentication, user definition ✗
- ☐ d. Access control, database security

The correct answer is: Access authentication, view definition

Question **51**

Correct

Mark 1.00 out of 1.00

11. Which of the following devices is layer 3(network Layer) device?

Select one:

- ☒ a. Router ✓
- ☐ b. Bridge
- ☐ c. Switch
- ☐ d. Hub

The correct answer is: Router

Question **52**

Correct

Mark 1.00 out of
1.00

13. _____ is refers to the objective an artificial machine.

Select one:

- ☐ a. Input
- ☐ b. Intermediate state
- ☐ c. Initial state
- ☒ d. Goal ✓

The correct answer is: Goal

Question **53**

Incorrect

Mark 0.00 out of
1.00

67. Which one of the following is not the advantage of using randomized algorithms?

Select one:

- ☐ a. Achieving worst case time-complexity.
- ☐ b. Efficiency in running time.
- ☒ c. Implementing problems requiring randomization. ✗
- ☐ d. Simplicity of implementation.

The correct answer is: Achieving worst case time-complexity.

Question **54**

Incorrect

Mark 0.00 out of
1.00

The following java code snippet shows a class definition containing only parametrized constructor. Based on the given code segment:

```
public class Car {  
  
    String type;  
  
    String model;  
  
    String color;  
  
    public Car(String type, String model, String color) {  
  
        this.type = type;  
  
        this.model = model;  
  
        this.color = color;  
  
    }  
  
}
```

Which of the following statement **wrongly** invokes the constructor?

Select one:

- ☒ a. Car golf = new Car ("Volkswagen", "Golf", "green"); ✖
- ☐ b. new Car ("Volkswagen", "Golf", "green");
- ☐ c.
Car focus = new Car ("Ford", "Focus", "red");
- ☐ d. Car auris = new Car ("Toyota", "Auris");

Your answer is incorrect.

The correct answer is: Car auris = new Car ("Toyota", "Auris");

Question **55**

Incorrect

Mark 0.00 out of
1.00

15. Which one of the following is different?

Select one:

- ☐ a. DFS
- ☐ b. BFS
- ☐ c. Uninformed searching techniques
- ☒ d. Informed Searching ✖

The correct answer is: DFS

Question **56**

Incorrect

Mark 0.00 out of 1.00

63. Which of these is not an objective for use-case modeling?

Select one:

- ☒ a. To provide a basis for validation testing ❌
- ☐ b. To define the functional and operational requirements
- ☐ c. To provide a description of end user and system interaction
- ☐ d. To define the hierarchy for the system

The correct answer is: To define the hierarchy for the system

Question **57**

Incorrect

Mark 0.00 out of 1.00

50. Which is the correct statement about operator overloading in C++?

Select one:

- ☐ a. Only arithmetic operators can be overloaded
- ☐ b. Only non-arithmetic operators can be overloaded
- ☒ c. Precedence of operators are changed after overloading ❌
- ☐ d. Associativity and precedence of operators does not change

The correct answer is: Associativity and precedence of operators does not change

Question **58**

Incorrect

Mark 0.00 out of 1.00

88. Various new methods of dealing with applets that works by dividing the virtual address space up into equal-size regions, which is called .

Select one:

- ☒ a. Trusted applet ❌
- ☐ b. Untrusted applet
- ☐ c. Sandboxes
- ☐ d. Interpreter

The correct answer is: Sandboxes

Question **59**

Incorrect

Mark 0.00 out of
1.00

87. One of the following is NOT the mechanism of the system threat that creates an environment to attack when operating system resources/ user files are misused, which one?

Select one:

- ☐ a. Trojan
- ☒ b. port scanning ✖
- ☐ c. Worm
- ☐ d. denial of service

The correct answer is: Trojan

Question **60**

Correct

Mark 1.00 out of
1.00

Suppose that you are given the following code snippet. Which one of the following statements is wrong about the given code snippet?

abstract class Animal {`public Animal (String name) {``System.out.println("I am animal class Constructor")``}`**abstract void** sound ();**final abstract void** eat ();**final void** move () {`//TODO``}``}`**public class** Lion **extends** Animal {**void** sound () {`// TODO``}`**public static void** main (String [] args) {`// TODO``}``}`

Select one:

- ☒ a. It is possible to override the **move ()** method in Lion class. ✔
- ☐ b. The **sound ()** method in the super class must be implemented in the subclass **Lion**
- ☐ c. The **eat()** method defined in **Animal** class is not supported
- ☐ d. It is must for the subclass to have at least one constructor.

Your answer is correct.

The correct answer is: It is possible to override the **move ()** method in Lion class.

Question **61**

Incorrect

Mark 0.00 out of 1.00

69. Among the following set of problems listed below, identify the one which can be solved using divide and conquer strategy?

Select one:

- ☐ a. Graph coloring problem
- ☐ b. Finding the maximum continuous sum in the array.
- ☒ c. N-queens problem. ✖
- ☐ d. Minimum spanning tree problem.

The correct answer is: Finding the maximum continuous sum in the array.

Question **62**

Correct

Mark 1.00 out of 1.00

48. Which of the following is true about FILE *fp

Select one:

- ☐ a. FILE is a keyword in C for representing files and fp is a variable of FILE type.
- ☐ b. FILE is a buffered stream
- ☒ c. FILE is a structure and fp is a pointer to the structure of FILE type ✔
- ☐ d. FILE is a stream

The correct answer is: FILE is a structure and fp is a pointer to the structure of FILE type

Question **63**

Incorrect

Mark 0.00 out of 1.00

49. What is an inline function?

Select one:

- ☐ a. A function that is expanded at each call during execution
- ☐ b. A function that is not checked for semantic analysis
- ☒ c. A function that is called during compile time ✖
- ☐ d. A function that is not checked for syntax errors

The correct answer is: A function that is expanded at each call during execution

Question **64**

Incorrect

Mark 0.00 out of 1.00

Consider the following grammar,

$$E \rightarrow E+T \mid T$$

$$T \rightarrow T*F \mid F$$

$$F \rightarrow \text{id} \mid (E)$$

which of the following statements are true about the parse tree constructed by deriving the string $\text{id}+\text{id}*\text{id}$ using left-most derivation?

Select one:

- ☒ a. The tree contains 3 terminal nodes ✖
- ☐ b. The height of the tree is 4
- ☐ c. The root node of the tree contains T
- ☐ d. The tree contains five internal nodes

Your answer is incorrect.

The correct answer is: The height of the tree is 4

Question **65**

Correct

Mark 1.00 out of 1.00

56. ____define the web contents and their structures, ____manage the layout of the web contents and their elements and ____control the interactivity of web site pages respectively.

Select one:

- ☐ a. CSS, HTML, XSL
- ☐ b. Javascript, VBscript and PHP
- ☐ c. HTML,CSS and XLS
- ☒ d. HTML, CSS and PHP ✔

The correct answer is: HTML, CSS and PHP

Question **66**

Incorrect

Mark 0.00 out of 1.00

66. While sorting the elements of the array in ascending order using quicksort, the worst-case time complexity will happen if the elements in the array have:

Select one:

- ☐ a. Arbitrary arrangement.
- ☒ b. Different values or arbitrary arrangement. ✖
- ☐ c. The same values or are already arranged in ascending order.
- ☐ d. Different values or are already arranged in descending order.

The correct answer is: The same values or are already arranged in ascending order.

Question **67**

Correct

Mark 1.00 out of 1.00

71. A linear collection of data elements where the linear node is given by means of pointer is called?

Select one:

- ☐ a. Node list
- ☐ b. Primitive list
- ☒ c. Linked list ✓
- ☐ d. Unordered list

The correct answer is: Linked list

Question **68**

Correct

Mark 1.00 out of 1.00

5. Which of the following concept of FSA is used in the compiler?

Select one:

- ☐ a. Code generation
- ☐ b. Parser
- ☐ c. Code optimization
- ☒ d. Lexical analysis ✓

The correct answer is: Lexical analysis

Question **69**

Incorrect

Mark 0.00 out of 1.00

45. Which one of the following RAID level offers optimal performance and reliability?

Select one:

- ☒ a. 1 ✗
- ☐ b. 0
- ☐ c. Any RAID level
- ☐ d. 0+1

The correct answer is: 0+1

Question **70**

Correct

Mark 1.00 out of 1.00

9. Which of the following IPv4 address belongs class B address?

Select one:

- ☐ a. 192.1668.1.248
- ☐ b. 224.1.120.29
- ☒ c. 172.12.12.48 ✓
- ☐ d. 121.12.12.8

The correct answer is: 172.12.12.48

Question **71**

Incorrect

Mark 0.00 out of 1.00

41. Suppose you are an ORACLE DBA and you need to create a trigger on the EMPLOYEES table that monitors every row that is changed and places this information into the AUDIT_TABLE. What type of trigger do you create?

Select one:

- ☐ a. FOR EACH ROW trigger on the AUDIT_TABLE table.
- ☒ b. Statement-level trigger on the EMPLOYEES table. ✗
- ☐ c. FOR EACH ROW trigger on the EMPLOYEES table.
- ☐ d. Statement-level trigger on the AUDIT_TABLE table.

The correct answer is: FOR EACH ROW trigger on the EMPLOYEES table.

Question **72**

Correct

Mark 1.00 out of 1.00

84. Sector 0 of the disk is called which is used to boot the computer.

Select one:

- ☒ a. MBR ✓
- ☐ b. Partition
- ☐ c. Bootblock
- ☐ d. Superblock

The correct answer is: MBR

Question **73**

Correct

Mark 1.00 out of 1.00

33. A register that manages the memory address of the instruction to be executed next.

Select one:

- ☐ a. Address register
- ☐ b. Accumulator
- ☐ c. data register
- ☒ d. Program counter ✓

The correct answer is: Program counter

Question **74**

Correct

Mark 1.00 out of 1.00

34. Information transfer from one register to another is designated in symbolic form by means of a replacement operator is:

Select one:

- ☐ a. Operation translation
- ☐ b. memory transfer
- ☒ c. Register transfer ✓
- ☐ d. common bus system

The correct answer is: Register transfer

Question **75**

Correct

Mark 1.00 out of 1.00

30. The structure and behavior of various functional modules (HW and SW) of a digital computer as well as how these modules interact to meet the requirement of the user.

Select one:

- ☐ a. computer specification
- ☐ b. computer application
- ☒ c. Computer architecture ✓
- ☐ d. Computer organization

The correct answer is: Computer architecture

Question **76**

Incorrect

Mark 0.00 out of 1.00

19. Which of the following is not a step in elimination of states procedure?

Select one:

- ☐ a. Unify single transitions to multi transitions that contains union of input
- ☐ b. Unifying all the final states into one using e-transitions
- ☐ c. Get the resulting regular expression by direct calculation
- ☒ d. Remove states until there is only starting and accepting states ✖

The correct answer is: Unify single transitions to multi transitions that contains union of input

Question **77**

Correct

Mark 1.00 out of 1.00

40. Which statement correctly differentiates functions and procedures?

Select one:

- ☒ a. A function must return a value to the calling environment, whereas a procedure can return zero or more values to its calling environment. ✔
- ☐ b. A function can be called only as part of a SQL statement, whereas a procedure can be called only as a PL/SQL statement.
- ☐ c. A function may return one or more values to the calling environment, whereas a procedure must return a single value to its calling environment.
- ☐ d. They only have syntax difference otherwise similar

The correct answer is: A function must return a value to the calling environment, whereas a procedure can return zero or more values to its calling environment.

Question **78**

Correct

Mark 1.00 out of 1.00

18. If we have more than one accepting states or an accepting state with an outdegree, which of the following actions will be taken?

Select one:

- ☐ a. removal of a state
- ☐ b. addition of new state
- ☒ c. more than one option is correct ✔
- ☐ d. make the newly added state as final

The correct answer is: more than one option is correct

Question **79**

Incorrect

Mark 0.00 out of 1.00

36. You create some new tables and populate them with copies of your production data. You run reports on your new tables but your queries take a lot longer than on the original tables. What should you create to help speed up reads in the database?

Select one:

- ☒ a. Primary key ❌
- ☐ b. Index
- ☐ c. Stored procedure
- ☐ d. Default

The correct answer is: Index

Question **80**

Correct

Mark 1.00 out of 1.00

61. Which types of system testing uses a procedure that demands resources in abnormal quantity, frequency or volume?

Select one:

- ☐ a. Unit testing
- ☐ b. Performance testing
- ☐ c. Security testing
- ☒ d. Stress testing ✔️
- ☐ e. Recovery testing

The correct answer is: Stress testing

Question **81**

Correct

Mark 1.00 out of 1.00

21. A language is regular if and only if

Select one:

- ☐ a. accepted by LBA
- ☐ b. accepted by PDA
- ☒ c. accepted by DFA ✔️
- ☐ d. accepted by Turing machine

The correct answer is: accepted by DFA

Question **82**

Incorrect

Mark 0.00 out of
1.00

76. 1. A Double-ended queue supports operations such as adding and removing items from both the sides of the queue. They support four operations like addFront(adding item to top of the queue), addRear(adding item to the bottom of the queue), removeFront(removing item from the top of the queue) and removeRear(removing item from the bottom of the queue). You are given only stacks to implement this data structure. You can implement only push and pop operations. What are the total number of stacks required for this operation?(you can reuse the stack)

Select one:

- ☐ a. 2
- ☒ b. 4 ✖
- ☐ c. 3
- ☐ d. 1

The correct answer is: 2

Question **83**

Incorrect

Mark 0.00 out of
1.00

65. The UML diagram that depicts the interaction between entities such as objects, classes, user interfaces etc participated in an execution of a certain use case is known as

Select one:

- ☐ a. Use case modeling
- ☐ b. Class diagram
- ☐ c. Sequence diagram
- ☒ d. State diagram ✖

The correct answer is: Use case modeling

Question **84**

Incorrect

Mark 0.00 out of
1.00

70. Which of the following is true about spanning trees?

Select one:

- ☒ a. A spanning tree will have $n-1$ vertices, if the original graph has n -number of vertices. ✖
- ☐ b. A spanning tree will have $n+1$ edge, if the original graph has n -number of vertices.
- ☐ c. A spanning tree will have n edges, if the original graph has n -number of vertices.
- ☐ d. A spanning tree will have $n-1$ edges, if the original graph has n -number of vertices.

The correct answer is: A spanning tree will have $n-1$ edges, if the original graph has n -number of vertices.

Question **85**

Correct

Mark 1.00 out of 1.00

29. The fundamental building block of all digital logic circuits.

Select one:

- ☒ a. gates ✓
- ☐ b. Adder
- ☐ c. Data
- ☐ d. Circuit

The correct answer is: gates

Question **86**

Incorrect

Mark 0.00 out of 1.00

39. Which one of the following statement is TRUE?

Select one:

- ☐ a. A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
- ☒ b. Every relation in 3NF is also in BCNF ✗
- ☐ c. No relation be both in BCNF & 3NF
- ☐ d. Every relation in BCNF is also in 3NF

The correct answer is: Every relation in BCNF is also in 3NF

Question **87**

Incorrect

Mark 0.00 out of 1.00

80. Which tool is used for managing and controlling network access and security policies on a Windows Server?

Select one:

- ☒ a. Network load balancing manager ✗
- ☐ b. Group policy management
- ☐ c. Organizational Unit
- ☐ d. Container

The correct answer is: Organizational Unit

Question **88**

Incorrect

Mark 0.00 out of
1.00

32. The most basic digital arithmetic circuit which Performs the addition of two binary digits.

Select one:

- ☒ a. decoder ✖
- ☐ b. multiplexer
- ☐ c. Full adder
- ☐ d. Half adder

The correct answer is: Half adder

Question **89**

Incorrect

Mark 0.00 out of
1.00

54. What is the preferred way for adding a background color in HTML?

Select one:

- ☒ a. <background color="yellow">text<background> ✖
- ☐ b. < body style="background-color:yellow">
- ☐ c. <body background="yellow">
- ☐ d. <background>yellow</background>

The correct answer is: < body style="background-color:yellow">

Question **90**

Correct

Mark 1.00 out of
1.00

62. Which system development method executes the phases of the SDLC in linear manner?

Select one:

- ☐ a. XP
- ☐ b. Scrum
- ☐ c. Prototyping
- ☒ d. Waterfall model ✔

The correct answer is: Waterfall model

Question 91

Correct

Mark 1.00 out of 1.00

14. Which one of the following is a correct representation of the fact "Abebe is father of Natan" using first order logic in prolog?

Select one:

- ☐ a. Father(abebe).
- ☐ b. Father(natan).
- ☒ c. father(abebe, natan). ✓
- ☐ d. black(abebe).

The correct answer is: father(abebe, natan).

Question 92

Incorrect

Mark 0.00 out of 1.00

37. Suppose there is a database that maintains business activities of the company. In the database Sales data is stored into two tables—one for recent sales (the past six months) and one for older or archived sales data. You are tasked with creating a solution to allow managers to be able to see monthly sales data from the previous month and for the same period a year ago. What would you do?

Select one:

- ☒ a. Create a trigger to identify when data is added and query the new data. ✗
- ☐ b. Create a view that joins data from both tables.
- ☐ c. Create indexes on the two tables based on the date and query the tables individually.
- ☐ d. Combine the data into a single table and query it.

The correct answer is: Create a view that joins data from both tables.

Question 93

Incorrect

Mark 0.00 out of 1.00

Given the following set of unbreakable items with their associated values in Birr and weights in kilograms in a table, what is the maximum profit that can be made having a knapsack with a capacity of 6 kg.

Value(Br.)	60	20	32	15	40	36
Weight(Kg.)	4	2	2	1	3	2

Select one:

- ☐ a. 96Br
- ☒ b. 88Br ✗
- ☐ c. 98Br
- ☐ d. 86Br

Your answer is incorrect.

The correct answer is: 96Br

Question **94**

Correct

Mark 1.00 out of 1.00

28. Which of the following is defined as an attempt to harm, damage or cause threat to a system or network?

Select one:

- ☒ a. Cyber Attack ✓
- ☐ b. Digital crime
- ☐ c. System hijacking
- ☐ d. Threats

The correct answer is: Cyber Attack

Question **95**

Incorrect

Mark 0.00 out of 1.00

24. Which of the following is an internet scam done by cyber-criminals where the user is convinced digitally to provide confidential information?

Select one:

- ☐ a. MiTM attack
- ☐ b. Phishing attack
- ☐ c. Website attack
- ☒ d. DoS attack ✗

The correct answer is: Phishing attack

Question **96**

Correct

Mark 1.00 out of 1.00

64. Polymorphism reduces the effort required to extend an object system by

Select one:

- ☐ a. Making objects more dependent on one another
- ☒ b. Enabling a number of different operations to share the same name ✓
- ☐ c. Coupling objects together more tightly
- ☐ d. Removing the barriers imposed by encapsulation

The correct answer is: Enabling a number of different operations to share the same name

Question **97**

Incorrect

Mark 0.00 out of
1.00

81. Which of the following backup technique is most space efficient?

Select one:

- ☐ a. RAID Technology
- ☐ b. Incremental backup
- ☐ c. Differential backup
- ☒ d. Full backup ✖

The correct answer is: Incremental backup

Question **98**

Incorrect

Mark 0.00 out of
1.00

43. User views are included as part of which schema?

Select one:

- ☐ a. External
- ☒ b. Conceptual ✖
- ☐ c. Internal
- ☐ d. Physical

The correct answer is: External

Question **99**

Incorrect

Mark 0.00 out of
1.00

90. Which one of the following is wrong about interface?

Select one:

- ☐ a. An interface can create an object of its own
- ☐ b. Interface is used to achieve full abstraction, loose coupling and multiple inheritance
- ☐ c. The default variable modifiers in an interface are public, static and final
- ☒ d. Methods in an interface are abstract, and public by default ✖

The correct answer is: An interface can create an object of its own

Question **100**

Incorrect

Mark 0.00 out of
1.00

17. The action of the Simple reflex agent completely depends upon _____.

Select one:

- ☐ a. Perception history
- ☐ b. Learning theory
- ☐ c. Utility functions
- ☒ d. Current perception ✖

The correct answer is: Learning theory